

AISI 1010 Steel, cold drawn**Categories:** [Metal](#); [Ferrous Metal](#); [Carbon Steel](#); [AISI 1000 Series Steel](#); [Low Carbon Steel](#)**Material Notes:** Used widely in low strength applications. Good formability, fair machinability, can be hardened by cyaniding. Suitable for magnet core applications**Key Words:** UNS G10100, AISI1010**Vendors:** No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
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Density	7.87 g/cc	0.284 lb/in ³	
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Mechanical Properties	Metric	English	Comments
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Hardness, Brinell	105	105	
Hardness, Knoop	123	123	Converted from Brinell
Hardness, Rockwell B	60	60	Converted from Brinell
Hardness, Vickers	108	108	Converted from Brinell
Tensile Strength, Ultimate	365 MPa	52900 psi	
Tensile Strength, Yield	305 MPa	44200 psi	
Elongation at Break	20 %	20 %	In 50 mm
Reduction of Area	40 %	40 %	
Modulus of Elasticity	205 GPa	29700 ksi	Typical for steel
Bulk Modulus	160 GPa	23200 ksi	Typical for steel
Poissons Ratio	0.29	0.29	Typical For Steel
Machinability	55 %	55 %	Based on AISI 1212 steel = 100%. Group I bar, rod, and wire products machinability can be improved by cold drawing.
Shear Modulus	80.0 GPa	11600 ksi	Typical for steel

Electrical Properties	Metric	English	Comments
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Electrical Resistivity	0.0000143 ohm-cm	0.0000143 ohm-cm	condition unknown
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Thermal Properties	Metric	English	Comments
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CTE, linear 	12.2 µm/m-°C @Temperature 0.000 - 100 °C	6.78 µin/in-°F @Temperature 32.0 - 212 °F	
	13.5 µm/m-°C @Temperature 0.000 - 300 °C	7.50 µin/in-°F @Temperature 32.0 - 572 °F	
	14.2 µm/m-°C @Temperature 0.000 - 500 °C	7.89 µin/in-°F @Temperature 32.0 - 932 °F	
Specific Heat Capacity 	0.448 J/g-°C @Temperature >=100 °C	0.107 BTU/lb-°F @Temperature >=212 °F	condition unknown
	0.498 J/g-°C @Temperature 150 - 200 °C	0.119 BTU/lb-°F @Temperature 302 - 392 °F	
	0.519 J/g-°C @Temperature 200 - 250 °C	0.124 BTU/lb-°F @Temperature 392 - 482 °F	

	0.536 J/g-°C @Temperature 250 - 300 °C	0.128 BTU/lb-°F @Temperature 482 - 572 °F	
	0.565 J/g-°C @Temperature 300 - 350 °C	0.135 BTU/lb-°F @Temperature 572 - 662 °F	
	0.590 J/g-°C @Temperature 350 - 400 °C	0.141 BTU/lb-°F @Temperature 662 - 752 °F	
	0.649 J/g-°C @Temperature 400 - 450 °C	0.155 BTU/lb-°F @Temperature 752 - 842 °F	
	0.729 J/g-°C @Temperature 550 - 600 °C	0.174 BTU/lb-°F @Temperature 1020 - 1110 °F	
	0.825 J/g-°C @Temperature 650 - 700 °C	0.197 BTU/lb-°F @Temperature 1200 - 1290 °F	
Thermal Conductivity	49.8 W/m-K	346 BTU-in/hr-ft²-°F	Typical steel

Component Elements Properties	Metric	English	Comments
Carbon, C	0.080 - 0.13 %	0.080 - 0.13 %	
Iron, Fe	99.18 - 99.62 %	99.18 - 99.62 %	As remainder
Manganese, Mn	0.30 - 0.60 %	0.30 - 0.60 %	
Phosphorus, P	<= 0.040 %	<= 0.040 %	
Sulfur, S	<= 0.050 %	<= 0.050 %	

[References](#) for this datasheet.

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.